

AD Project Risk: Get a Grip

Chartering risky AD projects with proper assessment of their risk can be strategically sound. But often risk is neither assessed or tracked, nor is potential impact on other projects evaluated.

Core Topics

Applications Development: Applications
Development Project Management

Key Issue

What strategies, processes and techniques will assist AD organizations in reducing their exposure to project failures?

Strategic Planning Assumptions

AD organizations that lack stringent risk assessment procedures will continue to cancel more than 20 percent of projects in the execution phase through 2002 (0.7 probability).

Note 1

Sample Questions: Project Size and Effort Risk

- What is the project's estimated duration?
- What is the project's estimated size in function points?
- What is the project's effort, in staff-hours?
- What is the estimated cost?
- Will multiple physical installations extend project implementation?
- How much will the project schedule depend on availability of end-user staff for analysis and testing?
- Will stringent quality requirements for the proposed system add to duration and cost for frequent inspections, documentation and testing?
- Is the end date fixed or flexible?

Project sponsors' business cases typically draw attention to all the benefits of their proposed projects, but cost/benefit analyses seldom seek to identify the risk to the overall portfolio that a project could pose. Danger signals around risky projects often go undetected well into a project's life, even as resource and duration requirements become clear. As a result, IS and AD organizations typically expend substantial resources on eventually canceled projects — wasted effort that is preventable with proper risk management, by which we do not mean managing schedule risk (e.g., via Monte Carlo-style simulation to anticipate critical path schedule slippage), but assessment of both project and portfolio risk. Any tardy or canceled project in the portfolio is comparable to a delay on the critical path of a project: A late project draining vital resources can throw off several other projects' plans. Therefore, prior to chartering a project, IS/AD management should make a disciplined risk assessment of any project likely to take more than a person-month, using a process similar to the following.

First, the project sponsor, working with the AD managers who may work on the eventual project, together detail the system's goals and specific requirements for the management group responsible for project chartering. Next, the chartering group, meeting or otherwise conferring monthly or quarterly, uses this information to rate the project from low- to high-risk (1 to 5) in three categories: 1) project size and effort; 2) business and requirements stability; and 3) resource capability (especially with new technologies or complex systems). Some sample questions to consider are given in Note 1, Note 2 and Note 3. IS/AD managers also regularly update the chartering group with key indicator information on the rest of the project portfolio, including chartered and prioritized projects and those actively in execution (see *Research Note SPA-04-7083*).

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Note 2

Sample Questions: Business and Requirements Risk

- Is the project in the IS/AD annual plan?
- Will business success depend on the project?
- Was cost of ownership analyzed?
- Do the intended users and their management accept and appreciate the value of the new system, or oppose it?
- How many departments or divisions are involved?
- Are there several complex deliverables?
- Will the project change whole business processes?
- How severely will user procedures change under the proposed system?
- Will company politics affect the project team's makeup?
- How large is the user population? How diverse?
- Will this be users' first experience with the type of system under development?
- Are the functional requirements clear, or vague?
- Will the system depend on many other business systems?
- Will the new system require new maintenance procedures?
- Could implementation interrupt mission-critical operations?
- Will user staff be available during development?
- Will the proposed system adhere to enterprise standards?

Note 3

Sample Questions: Resource and Skill Risk:

- Are more than three distinct systems involved?
- Are critical tasks out of the project team's control?
- Will the project require major hardware or software upgrades?
- Must multiple IS/AD departments provide resources to the project?
- Will the IS/AD staff be continuously available throughout the project?
- Is any software (i.e., language, database, communications or tools) for the project new to the development team?
- Is any hardware new to the development team?
- Will construction require complex and intricate logic?
- Will project teams inexperienced in the product or business areas be on the project?

Project effort estimates prepared for the business case ought to be based on systematic estimating techniques; a "large" rating of 5 might be for a project estimated at more than 30,000 staff hours. The most accurate estimating combines three approaches: 1) experience-based estimates (if personnel experienced with the business and technical problems are available); 2) task/deliverable estimates based on a methodology route that breaks out the necessary tasks or deliverables; and 3) a function-point size estimate sometimes supplemented with a database of analogous projects (see *Strategic Analysis Report R-930-140*, 4 November 1996, for a discussion of estimating techniques and tools).

The second rating category looks at stability of the project in terms of its business requirements and business environment (see Note 1, Note 2 and Note 3). The third category, resource risk, refers mainly to resources' skill and experience with the requisite technologies, and to project managers' experience, especially on similar projects. Only the skills and experience of the team of people likely to actually work on the project are relevant. One benefit of tool-enabled portfolio management is the ability to efficiently move skilled resources among projects and track key skills and experience gained.

Analysis of feedback gathered during the "assessment" portion of the chartering phase of our recommended portfolio validation process will ultimately tell the AD organization what scores indicate danger. Initially, an average score of greater than 3 might result in rejection or redefinition of the project (e.g., via a more compelling business case, stricter project controls or more frequent deliverables) to reduce risk. Proposed projects should also be subject to our recommended five-step viability review (see *Research Note SPA-03-1250*). Using the viability review and risk assessment together can make a project's first gate a difficult one to pass, and ensure a valid project portfolio.

Bottom Line: IS and AD management should conduct stringent risk assessment of any significant project before deciding whether or not to charter it. AD organizations that lack stringent risk assessment procedures will continue to cancel more than 20 percent of projects in the execution phase through 2002 (0.7 probability).

